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Combining predictive distributions for time-to-event outcomes in meteorology

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ABSTRACT

Combining forecasts from multiple numerical weather prediction (NWP) models has shown substantial benefit over the use of individual forecast products. Although combination, in a broad sense, is widely used in meteorological forecasting, systematic studies of combination methodology in meteorology are scarce. In this article, we study several combination methods, both state-of-the-art and of our own making, with a particular emphasis on situations where one seeks to predict *when* a particular event of interest will occur. Such time-to-event forecasts require particular methodology and care. We conduct a careful comparison of the different combination methods through an extensive simulation study, where we investigate the conditions under which the combined forecast will outperform the individual forecasting products. Further, we investigate the performance of the methods in a case study modelling the time to the first hard freeze in Norway and parts of Fennoscandia.

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1. Introduction

In many real forecasting situations, one may have access to multiple forecasts based on different sources of information.[1] In meteorology, for example, combining predictions from several numerical weather prediction (NWP) models frequently results in better predictions than any of the single models. In this article, we investigate how best to combine probabilistic weather forecasts, with a specific focus on time-to-event prediction. We also attempt to clarify, through simulations and a case study, under what conditions a combination will be fruitful, i.e. that the combined forecast is more skilful than any of the single-source forecasts.

Combinations of weather forecasts from different sources have been studied by several authors; see for instance (Baran & Lerch, 2018; Gneiting & Ranjan, 2013; Kober et al., 2012; Schaumann et al., 2020; Vislocky & Fritsch, 1995). The terms 'combination', 'aggregation', 'merging', and 'blending' are used more or less interchangeably. Older works combine the forecasts by taking simple linear averages of the different forecast distributions (Kober et al., 2012; Vislocky & Fritsch, 1995), often giving equal weights to the different forecasts. Later contributions have explored more complex combination methods and typically let combination weights, and sometimes other parameters, be estimated from historical data (Baran & Lerch, 2018; Gneiting & Ranjan, 2013; Schaumann et al., 2020). There are similarities between these combination techniques and techniques used for statistical post-processing of weather forecasts; see for example

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the BMA approach in Raftery et al. (2005) and the EMOS approach in Gneiting et al. (2005). In the field of econometrics, many similar methods have been developed and studied in parallel; see Wang et al. (2022) for a recent review.

Meteorological forecasts are different from forecasts in most other fields, in terms of both their origin and the form they are issued in. Unlike fields where forecasts are the result of statistical models on historical data, weather forecasts are produced from NWP models, i.e. physical models of the atmosphere and oceans (Baran & Lerch, 2018).

Further, meteorological forecasts usually come in the form of ensembles, where each ensemble member corresponds to a run of the model with different initial conditions and model physics (Baran & Lerch, 2018). The different ensemble members are typically considered exchangeable in the statistical sense, and the ensemble may be interpreted as a random sample from the underlying forecasting distribution. In this article, ensembles from different types of NWP models constitute the different forecasts which we want to combine. This could for example be ensembles from different weather centres, forecasts with different release frequencies and lead times, or a mix of dynamic and statistical forecasts. Our study focuses on the combination of seasonal and subseasonal forecasts.

Seasonal forecasts provide forecasts of weather conditions up to five to seven months ahead, while subseasonal forecasts provide weather information for the next two to three months. Recent studies from Europe show that subseasonal and seasonal forecasts often have skill the first three to four weeks, particularly for temperature-based variables (Baker et al., 2023; Ceglar & Toreti, 2021; Hyvärinen et al., 2020; Roksvåg T. Lenkoski et al., 2023), and occasionally skill for even longer lead times (Vajda & Hyvärinen, 2020).

Many users want to use meteorological forecasts to anticipate *when* a particular event of interest will occur. For example, drought predictions (Yuan & Wood, 2013), predictions of the onset of the next rainy season (Scheuerer et al., 2024), predictions of the time-to-hard-freeze in agriculture (Roksvåg T. Lenkoski et al., 2023), and predictions of the flowering time for wheat Ceglar and Toreti (2021). Time-to-event outcomes such as these should be handled with methods from survival analysis. This subfield has deep roots in medical statistics, but is also frequently used in a diverse set of fields ranging from engineering to economics. Survival analysis methods are especially well suited for handling censored observations, which in the meteorological context occur when the target event does not occur before the forecast's maximum lead time. To our knowledge, Roksvåg T. Lenkoski et al. (2023) is the first instance of the use of survival time methods in meteorology. Roksvåg T. Lenkoski et al. (2023) employed survival time methods based on the output of a single NWP model. Our present work extends this methodology to consider how forecasts from multiple sources can be combined effectively.

Despite the relatively extensive literature on the topic of forecast combination, there are only a few systematic studies and comparisons of different combination methods in the context of meteorological forecasting. Baran and Lerch (2018) is an exception. They compared several forecast combination methods, specifically focusing on combining two statistical distributions fitted to the same forecast ensemble in the context of post-processing. In their conclusion, they highlighted that the combination methods they studied may also be used to combine different forecasts. This is what we address in the present paper. Through simulation experiments and a real data case study, we provide a thorough study of combination methods, particularly relevant to the combination of forecasts in meteorology. Our case study and simulation experiment are time-to-event setups, but the conclusions we draw about the combination methods should generalise to other outcomes in meteorology.

While our work is motivated by applications in meteorology, the approach to merging multiple survival time estimates draws directly on the broader statistical literature in distributional forecast combination (Gneiting & Raftery, 2007). A survival time estimate is inherently distributional in nature. Indeed, in situations without censoring, our framework has strong similarities to that considered by Gneiting and Ranjan (2013). Our work therefore builds upon the existing statistical forecasting literature for probabilities while extending its application to situations where constituent forecast systems can have a different temporal extent.

Our contribution in this paper is (1) a careful comparison of different combination methods, through an extensive simulation study and a real, interesting application; (2) two new combination methods; and (3) the application of (more) survival analysis tools to the weather prediction context.

In the next section, we present the real data application which serves as an illustration and case study throughout the paper. Then in Section 3, we describe the methods that will be used: the necessary survival analysis methodology, as well as the combination methods and the framework for forecast evaluation. In Section 4, we return to the application to demonstrate the use of the combination methodology in a small example. Section 5 contains the simulation study, and Section 6 the case study. Finally, we discuss our findings and conclude in Section 7.

2. Motivating the case study

Throughout the paper, we consider a particular forecasting application, which we describe here and return to in Sections 4 and 6. The application also served as inspiration for the design of the simulation study in Section 5. Our goal in the case study was to predict the time to the first hard freeze of the season, subsequently referred to as the time-to-hard-freeze. A hard freeze is defined as a mean daily near-surface air temperature below 0 °C. Such definitions should be made with the forecast user in mind. Here, for example, the definition of a hard freeze was coordinated with Norwegian farmers, as our study locations are in Norway and parts of Fennoscandia. In this region, the first hard freeze of the season typically comes sometime after October 1.

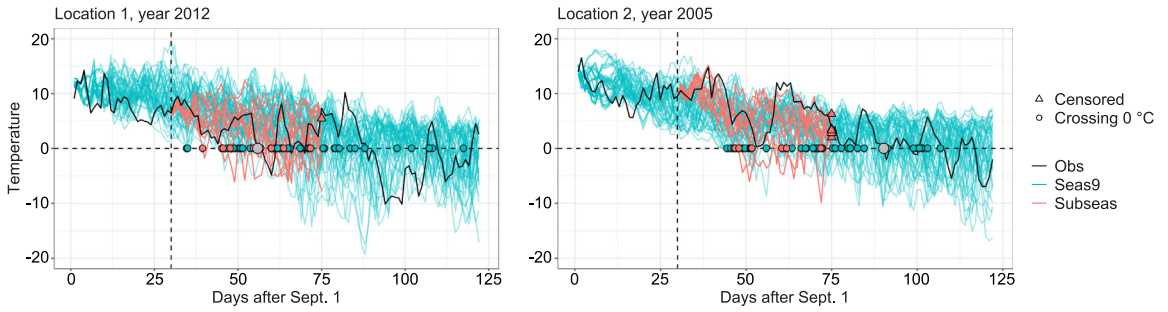


Fig. 1. Temperature trajectories for ensemble members of the seasonal forecast (Seas9), issued September 1, and the subseasonal forecast (Subseas), issued September 30, for two different locations in Fennoscandia for two different years. The observed temperature is shown in black (Obs). The first time each temperature trajectory crosses 0 °C is marked with a circle. Members that never cross zero before the maximum lead time is reached are marked with a triangle and are treated as censored in the survival analysis framework.

Two ensemble forecast products for surface temperature with forecasts covering the relevant time period are available:

1. the seasonal forecast: an ensemble of around 50 members with an issue date of September 1 and a maximal lead time of 122 days (December 31); and
2. the subseasonal forecast: an ensemble of 11 members with an issue date of September 30 and a maximal lead time of 46 days (November 15).

The two forecasting products are referred to as our two sources of information in the following. The seasonal forecasts we use here were also considered by Roksvåg T. Lenkoski et al. (2023), where the authors demonstrated that seasonal forecasts alongside survival analysis methods can produce skilful forecasts of the time-to-hard-freeze. The question is now: Can we obtain an even more skilful forecast by combining the seasonal and subseasonal forecasts?

Fig. 1 provides the observed temperature and forecasted temperature trajectories over the autumn season in two different years (and locations). Each coloured line represents the forecasted temperature trajectory from a particular ensemble member.

From each forecasted temperature trajectory, we obtain a forecast for the time-to-hard-freeze by recording the time when the trajectory first crosses 0 °C. This results in an ensemble of time-to-hard-freeze forecasts from each of the two information sources. These time-to-event forecasts are shown as blue (seasonal) or red (subseasonal) dots in the figure. The actual observed time-to-hard-freeze is given as a grey dot. The figure illustrates that some trajectories will reach their maximal lead time without crossing the 0 °C line. In the figure, these instances are indicated by red triangles, and these time-to-event observations are considered *censored*. The exact value of a censored time-to-event is not known; we only know that the event of interest will happen at some point after the censoring date. Survival analysis methods are specifically designed to account for censored observations, ensuring accurate analysis even when the timing of the event of interest has not been fully observed.

In the two examples shown in Fig. 1, we only have censored observations in the subseasonal ensemble. This will often be the case in our case study as well, due to the short lead time of the subseasonal forecast. Note however that we can have censoring in the seasonal ensemble too, and for some particular combinations of the year and location, we can even have censored realisations, meaning that the hard freeze does not occur before December 31 (and the black line in the plots above does not cross below 0 °C).

In our case study, we assume that we are on October 1, and that we want to predict the time-to-hard-freeze, if the hard freeze has not happened already. This date is indicated by a vertical dashed line in Fig. 1. The date was chosen out of convenience, and because hard freezes happen after October 1 for most of the 103 study locations we treat in the case study (see Section 6). On October 1, the seasonal forecast is already old (having been issued on September 1), but as we see below, it still has some predictive skill, at least when combined with the subseasonal forecast.

The data sources for the observations and forecasts are described in detail in Appendix D.1.

3. Methods

We denote the variable we want to forecast as T, indicating the time to the event, for example, the time-to-hard-freeze in our motivating application. The combination methods we study require estimation in two steps. In the first step, each source is analysed separately to produce source-specific forecast distributions that account for censored observations. In the second step, these source-specific forecast distributions are combined. This step typically requires the estimation of combination weights, and potentially other combination parameters. In the following section, we start by presenting the methods we use in the first step. These are standard methods from the field of survival analysis. Then, in Section 3.2 we present the combination methods, most of which have utility in contexts beyond time-to-event studies. We note that our selection of forecast combinations is quite broad, and we attempted to include many of the main forecast combination paradigms in the literature.

3.1. Single-source analyses

Our information sources are meteorological forecasting products which come in the form of ensembles. So

for a particular forecasting situation (i.e. a particular time and place) and a specific information source k, we have access to a sample of n_k time-to-event forecasts, which we use to estimate a source-specific forecasting distribution $\widehat{F}_k(t)$ in the form of a cumulative distribution function (CDF). The hat highlights that $F_k(t)$ is estimated from the ensemble members, but we sometimes omit the hats in the following for ease of notation. In survival analysis, it is typical to focus on the survival function $S_k(t)$ instead of the CDF, i.e. $S_k(t) = 1 - F_k(t)$, and we adhere to that convention in the following. The survival function gives the probability that the event of interest, e.g. the hard freeze, has not occurred before time t.

The survival function $S_k(t)$ can be estimated using either non-parametric or parametric tools. We present some of these time-to-event tools here. The observations in the ensemble of typical survival data form $(t_1, \delta_1), \ldots, (t_{n_k}, \delta_{n_k})$, where t_i is the possibly censored time-to-event, and δ_i is the censoring indicator, with $\delta_i = 1$ indicating that the event occurred at time t_i, and $\delta_i = 0$ indicating that the observation was censored at time t_i.

The Kaplan–Meier (KM) estimator (Kaplan & Meier, 1958) is the standard non-parametric estimator of the survival function:

$$\widehat{S}_k(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right) \tag{1}$$

where t_i denotes the ordered event time, d_i is the number of events at time t_i, and n_i is the number of individuals at risk just before time t_i. Censored members are not counted in the numerator d_i, as they have not experienced the target event, but they are counted in the denominator n_i, as members at risk before the censoring time. The KM estimator provides a step-function estimate of the survival probability over time.

Alternatively, one may use parametric methods to estimate $S_k(t)$. In this paper, we only consider log-normal models, but there are of course many other options. The log-normal is typically parameterised by a mean ξ and a variance parameter τ^2 on the log-transformed scale. The CDF of the log-normal may be written as

$$F_k(t) = \Phi\left(\frac{\log(t) - \xi}{\tau}\right), \tag{2}$$

where $\Phi(\cdot)$ is the standard normal CDF. We estimate the source-specific parameters (ξ, τ) by either maximum likelihood or by minimizing the integrated Brier score (see Eq. (8)). In either case, we obtain a continuous $\widehat{S}_k(t)$ by plugging in the estimated parameters into Eq. (2), above. See Appendix C for a slightly more advanced method to get $\widehat{S}_k(t)$ from the estimated log-normal model.

In a time-to-event setup, the log likelihood takes the following form:

$$\ell_{n_k}(\xi, \tau) = \sum_{\delta_i=0} \log S_k(t; \xi, \tau) + \sum_{\delta_i=1} \log f_k(t; \xi, \tau), \tag{3}$$

where the first part consists of a sum over the censored observations, while the second part is over the non-censored observations. In our case, $S(t; \xi, \tau)$ is the log-normal survival function and $f(t; \xi, \tau)$ the log-normal density. The density $f_k(t; \xi, \tau)$ represents the probability that the target event occurs at the exact observed time, whereas $S_k(t; \xi, \tau)$ captures the probability that the event takes place some time after the censoring time. The above likelihood formulation allows us to estimate (ξ, τ) taking into account the censored observations.

3.2. Combination methods

Here, we present methods for the second step of the combination procedure, where the source-specific forecasting distributions are combined to obtain a combined forecasting distribution, $S_c(t)$. We present four combination methods, which we denote by LP, BP, GP, and HB (acronyms explained below). The first two, LP and BP, are state-of-the-art methods in the literature, while the next two are our own contributions. We present the methods for the case where one has two sources of information, but it is straightforward to generalise to more than two sources. From the source-specific analyses in the first step (Section 3.1), we obtain two estimated survival functions, $\widehat{S}_1(t)$ and $\widehat{S}_2(t)$, which may be either step functions or continuous, as we saw above. From these, we easily have the estimated CDFs $\widehat{F}_1(t)$ and $\widehat{F}_2(t)$, which we will use in the combination step. Note that we omit the hats on the single-source CDFs below for ease of notation. For all combination methods except HB, censoring among the ensemble members is taken into account in the estimation of the source-specific forecast distributions; see Section 3.1.

3.2.1. Linear pooling (LP)

Linear pooling combines $F_1(t)$ and $F_2(t)$ by a taking a simple weighted sum:

$$S_c^{\mathrm{LP}}(t) = 1 - [\omega F_1(t) + (1 - \omega) F_2(t)]. \tag{4}$$

The combined survival curve $S_c^{\mathrm{LP}}(t)$ depends on a weight ω, which reflects the quality of source 1 relative to source 2. The parameter ω is usually estimated from training data, i.e. past forecast–observation pairs. This estimation step is explained in Section 3.4. Sometimes it is instead simply set to $\omega = 0.5$.

Gneiting and Ranjan (2013) studied this combination method in some detail, and proved that if the two single-source forecasts are calibrated, then the combined forecast $S_c(t)$ will be overdispersed. Thus, $S_c(t)$ is not calibrated, and this is a consequence of the LP failing to be flexibly dispersive, using the terminology of Gneiting and Ranjan (2013).

The linear pool method is in widespread use in econometrics; see Aastveit et al. (2018) and Hall and Mitchell (2007). It has also been used in meteorological contexts; see for instance Baran and Lerch (2018) and Schaumann et al. (2020).

There exists an extension to the simple LP method with one additional parameter, the spread-adjusted linear pool (SLP). The additional parameter aims at adjusting the variance of the single-source forecasts, but in contrast to the BP and GP methods described next, the SLP method is not flexibly dispersive (Gneiting & Ranjan, 2013). Therefore, we did not consider this method in detail, although we return to it briefly in the case study in Section 6.

3.2.2. Beta pooling (BP)

Gneiting and Ranjan (2013) proposed the beta-transformed linear pool, which we simply call the beta pooling (BP) method. Here, the single-source CDFs are combined by the following formula:

$$S_c^{\mathrm{BP}}(t) = 1 - B_{\alpha,\beta}[\omega F_1(t) + (1-\omega)F_2(t)]. \tag{5}$$

where $B_{\alpha,\beta}(\cdot)$ is the beta CDF, with parameters α and β. Note that if $\alpha = \beta = 1$, we have $S_c^{\mathrm{BP}}(t) = S_c^{\mathrm{LP}}(t)$, because in that case the beta distribution is equal to a uniform one. The combination parameters (α, β, ω) must be estimated from training data; see Section 3.4. Gneiting and Ranjan (2013) proved that the BP is flexibly dispersive. It therefore produces a combined forecast $S_c^{\mathrm{BP}}(t)$ that is calibrated, even when both single-source forecasts are also calibrated.

The BP is in less widespread use than LP, but is for example investigated by Baran and Lerch (2018), and extended by Bassetti et al. (2018).

3.2.3. Gaussian pooling (GP)

Gaussian pooling (GP) combines the single-source CDFs in the following way:

$$S_c^{\mathrm{GP}}(t) = 1 - \Phi\left[\frac{\omega\Phi^{-1}\{F_1(t)\} + (1-\omega)\Phi^{-1}\{F_2(t)\} - \mu}{\sigma}\right]. \tag{6}$$

where ω is the weight given to source 1, and $\Phi(\cdot)$ and $\Phi^{-1}(\cdot)$ are the standard normal CDF and quantile function, respectively. Here, we have three combination parameters (μ, σ, ω), which again must be estimated from training data. The role of the combination parameter is investigated in the illustration in Section 4 and in the simulation study.

A version of this method, without the parameters μ and σ, has been considered in other parts of the literature, for example very briefly by Gneiting and Ranjan (2013) under the umbrella of generalised linear combination formulas. Similar to the BP method, the three-parameter GP is flexibly dispersive; see Theorem 1 in the Appendix. We find that the combination parameters in the GP have a more straightforward interpretation than the combination parameters in BP.

Further, GP has a very natural extension to situations where the training dataset is small. Then, it is natural to use the following Student-t version of GP, which we will call GP-t:

$$S_c^{\mathrm{GP-t}}(t) = 1 - G_{n-1}\left[\frac{\omega\Phi^{-1}\{F_1(t)\} + (1-\omega)\Phi^{-1}\{F_2(t)\} - \mu}{\sigma}\right], \tag{7}$$

where $G_{n-1}(\cdot)$ is the CDF of the Student-t distribution with $n-1$ degrees of freedom, and n is the number of training samples.

3.2.4. Hazard blending (HB)

The previous three combination methods combine the CDFs from the two sources, while hazard blending (HB) works on the hazard scale:

$$S_c^{\mathrm{HB}}(t) = \prod_{t_i \le t}(1-\lambda_i), \text{ with } \lambda_i = \frac{\omega d_{1,i} + (1-\omega)d_{2,i}}{\omega n_{1,i} + (1-\omega)n_{2,i}}.$$

Here, $d_{1,i}$ is the number of ensemble members from source 1 for which the event of interest occurred at time t_i, while $n_{1,i}$ is the number of ensemble members from source 1 for which the event has not yet occurred at time t_i. The numbers $d_{2,i}$ and $n_{2,i}$ have similar roles for source 2.

This combination method is inspired by the long-established Kaplan–Meier estimator, see Eq. (1), which is the standard non-parametric estimator of a survival function in the presence of censored observations. HB is thus a non-parametric combination method, which does not require any parametric model for F_1 and F_2, and handles censoring similarly to the KM estimator. HB always results in a survival function $S_c^{\mathrm{HB}}(t)$ in the form of a step-function, while the three other combination methods may result in either step functions or continuous functions depending on the analyses in the first step of the combination procedure (Section 3.1).

We refer to hazard blending (HB) and linear pooling (LP) as simple combination methods, in the sense that they only have a single combination parameter ω. The ω in HB and LP cannot be compared directly, however, since they belong to different scales. In contrast to LP and HB, we refer to beta pooling (BP) and Gaussian pooling (GP) as complex combination methods, since they have three combination parameters each.

3.3. Forecast evaluation

In our simulation study and in the real data application, we evaluate the forecasts, either from single sources or from combinations, with respect to sharpness under calibration, as is commonly done (Gneiting et al., 2007). Calibration is assessed using the probability integral transform (PIT) histograms (see for instance Thorarinsdottir and Schuhen (2018)), and we compare the mean and standard deviation of the PIT values with 0.5 and $1/\sqrt{12}$, respectively, which should be their values under perfect calibration.

Predictive performance is assessed by the integrated Brier score (IBS). If we have m realised times to the event on which to evaluate our forecasts, the Brier score (Brier, 1950) takes the form

$$\mathrm{BS}(t) = \frac{1}{m}\sum_{i=1}^{m}[\mathbb{I}\{T_i > t\} - S_i(t)]^2,$$

where $S_i(t)$ is the forecast which we want to evaluate, and T_i is the observed time-to-hard-freeze. The Brier score is a function of time t, so in order to obtain a single numerical value, we integrate the score over time t. In practice, we follow Roksvåg T. Lenkoski et al. (2023) and discretise the timespan into days and sum from one to $t_{\max}$, the maximum number of days ahead that we consider:

$$\mathrm{IBS} = \sum_{t=1}^{t_{\max}}\mathrm{BS}(t) = \sum_{t=1}^{t_{\max}}\frac{1}{m}\sum_{i=1}^{m}[\mathbb{I}\{t_i > t\} - S_i(t)]^2. \tag{8}$$

In the case study, we use $t_{\max} = 120$. The IBS is equivalent to the continuous ranked probability score (CRPS), which is typically used when assessing the quality of predictive distributions. We chose to use the term 'IBS' to describe this scoring system, as it highlights that in the case of assessing survival distributions, the CRPS is the generalization of the BS, which would be used for assessing individual probabilities.

3.4. Estimation of combination parameters

The combination methods contain combination parameters, the weight ω and potentially others, which are typically estimated from a number of available forecast–observation pairs. For ease of exposition, we assume here that we have these training data from a number of years at a single location. The training dataset then consists of (1) n past time-to-event realisations at the particular location ($t_i : i = 1, \ldots, n$), and (2) past ensemble forecasts from each of the two sources for each year i. We consider two estimation methods, one based on maximising the likelihood and the other on minimizing the IBS.

Note that in our case study and also in other time-to-event applications, we may have censored realisations, in addition to the censored ensemble members (described in Section 3.1). These censored realisations need to be accounted for in the estimation of the combination parameters. Specifically, the log likelihoods below need to be amended in a similar way as the log likelihood in Eq. (3).

3.4.1. Maximum likelihood (ML)

The parameters of the combination methods LP, BP, and GP can be estimated by maximising the likelihood of the combination formula over the training data. For example, this may look like the following for LP:

$$\ell(\omega) = \sum_{i=1}^{n} \log[\omega f_{1,i}(t_i) + (1-\omega) f_{2,i}(t_i)], \tag{9}$$

where $f_{1,i}(\cdot)$ and $f_{2,i}(\cdot)$ are the forecast densities for year i from the two sources.

For the BP method, the log likelihood can be found in Gneiting and Ranjan (2013) or by differentiating Eq. (5). For the GP method, differentiating Eq. (6) results in the following log likelihood:

$$\begin{aligned}\ell(\omega, \mu, \sigma) = \sum_{i=1}^{n} \log\{\phi\left(\frac{\omega\Phi^{-1}\{F_{1,i}(t_i)\} + (1-\omega)\Phi^{-1}\{F_{2,i}(t_i)\} - \mu}{\sigma}\right) \\ \frac{1}{\sigma}[\omega\Phi^{-1\prime}\{F_{1,i}(t_i)\} f_{1,i}(t_i) \\ + (1-\omega)\Phi^{-1\prime}\{F_{2,i}(t_i)\} f_{2,i}(t_i)]\},\end{aligned}$$

where $\phi(\cdot)$ is the standard normal density function, $\Phi^{-1\prime}(\cdot)$ denotes the derivative of the standard normal quantile function (which may not have an analytical expression but is simple to evaluate numerically). $F_{1,i}(\cdot)$ and $F_{2,i}(\cdot)$ are the forecast CDFs, and $f_{1,i}(\cdot)$ and $f_{2,i}(\cdot)$ the forecast densities for year i for each of the two sources. In the context of combination, this estimation method is used by Gneiting and Ranjan (2013) and Baran and Lerch (2018), for example.

3.4.2. Minimising the IBS

Instead of ML, one may find the combination parameters by numerically optimising other loss functions, for instance the IBS. For example, for the HB method, we minimise

$$\text{IBS}(\omega) = \sum_t \frac{1}{n} \sum_{i=1}^{n} (\mathbb{I}\{t_i > t\} - S_{c,i}^{\text{HB}}(t))^2,$$

with respect to ω. Here, $S_{c,i}^{\text{HB}}(t)$ is the predictive survival curve from year i, and t_i is the realised time-to-hard-freeze in year i. This method may be used for LP, GP, and BP, too. This estimation method was used in the context of combination by Baran and Lerch (2018).

4. Illustrating combination

For concreteness, we show the combination methods at work in a specific example. Assume we are in the location of Torsby in Sweden on October 1, 2018, and we want to predict the time-to-hard-freeze. As explained in Section 2, we have two forecasting products which will be our two sources of information: the seasonal and subseasonal forecasts.

These are the same data we use in the case study in Section 6, but here, we only look at one specific location.

We have 19 forecast–observation pairs for the years between 1999 and 2017. These are our training data, from which we estimate combination parameters. In Fig. 2 , we see the forecasts from the two sources for some of the training years. In some years, the two sources are similar. In other years, they differ substantially. Sometimes the seasonal forecast is better (for example, in 2010), while for other years the subseasonal is closest to the realised time-to-hard-freeze (for example, in 2011). We see that some of the ensemble members in the subseasonal forecast are censored in 2010: three of the 11 members predict that frost will come after day 46, i.e. some time after mid-November, beyond the lead time of the subseasonal forecast.

In the first step of the combination procedure, we fit parametric log-normal models to the single-source forecasts in each year, as described in Section 3.1. The resulting continuous survival curves are used in the next step of the combination procedure by some of the combination methods (LP, BP and GP in this case). These parametric survival curves are shown as smooth lines in Fig. 2, alongside the step functions, which are the KM estimators for each of the two sources. We see that the log-normal models fit the ensemble data reasonably well.

In the second step of the combination methods, we estimate the combination parameters based on the 19 years of training data. Here, we use maximum likelihood for LP, BP, and GP, and minIBS for HB. We obtain the following estimated combination parameters:

$$\begin{aligned}&\widehat{\omega}_{\text{LP}} = 0.49 \quad \text{(weight on seasonal)} \\ &\widehat{\omega}_{\text{BP}} = 0.49 \quad \hat{\alpha} = 1.63 \quad \hat{\beta} = 1.21 \\ &\widehat{\omega}_{\text{GP}} = 0.79 \quad \hat{\mu} = 0.22 \quad \hat{\sigma} = 0.92 \\ &\widehat{\omega}_{\text{HB}} = 0.09\end{aligned}$$

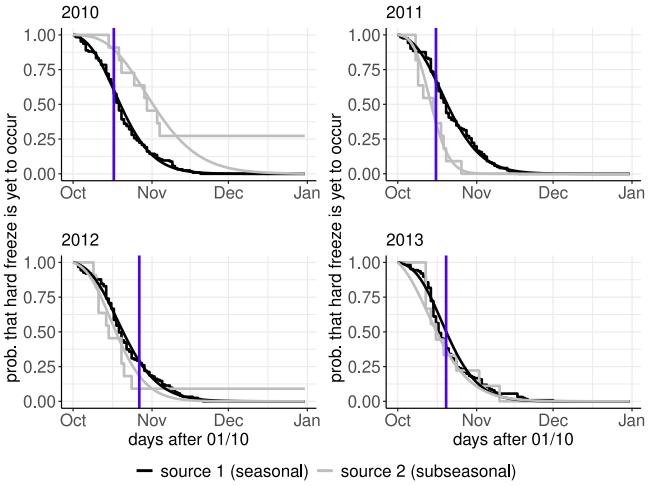


Fig. 2. Four years of training data (past forecast–observation pairs). The vertical blue line marks the realised time-to-hard-freeze in each year. The grey and black step functions are the KM estimators for the subseasonal forecast (grey) and seasonal forecast (black). The smooth functions in the same colours are log-normal models fitted to these ensembles. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

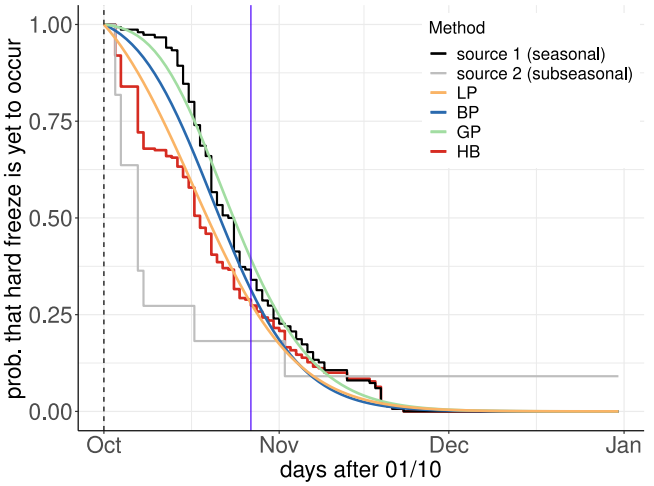


Fig. 3. Forecasts for 2018. The grey and black step functions are the KM estimators for the subseasonal forecast (grey) and seasonal forecast (black). The coloured survival curves are the resulting predictive survival curves from the four combination methods. The vertical blue line marks the realised time-to-hard-freeze. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Note again that ω for LP and HB are on different scales. The one from LP is more directly interpretable and indicates that the two sources are considered approximately equally good, on average, for this location in the years 1999 to 2017. The additional combination parameter in the GP method, μ and σ, can correct potential biases and over- or under-dispersion in the combined forecast. Here μ is relatively close to 0, so there does not seem to be much bias in the combined forecast. The standard deviation σ is a bit below 1, which is a typical value when both single-source forecasts are close to calibrated. The parameters α and β in BP play a similar role as μ and σ, but are a bit less directly interpretable. The further they are from (1, 1), the more they correct the mean and variance of the combined forecast.

In Fig. 3, we see that the predictive survival curves from the combination methods tend to lie between the survival curves of source 1 and source 2. This will typically be the case, but for BP and GP, the combined survival curve can lie outside the two single-source survival curves—in cases where there is bias in one or both sources. HB produces a step function. The others produce smooth curves, since we use the parametric single-source survival curves in this case. LP and HB are otherwise quite similar, and that is typical in our experience. Compared to the other methods, GP is closer to the seasonal here—as

Table 1
Probabilistic forecasts compared in the simulation experiment. Source-specific estimation can be either parametric (ML) or non-parametric. Combination estimation can be by ML or by minimising the IBS. The last column indicates the number of combination parameters.

Method	Source-specific est.	Combination est.	No. params.
source 1	ML (log-normal)	–	–
source 2	ML (log-normal)	–	–
source 1	KM (non-parametric)	–	–
source 2	KM (non-parametric)	–	–
LP	ML (log-normal)	ML	**1**
BP_3	ML (log-normal)	ML	**3**
GP_3	ML (log-normal)	ML	**3**
HB	**non-parametric**	**min IBS**	**1**
LP_{IBS}	ML (log-normal)	min IBS	1
BP_{IBS}	ML (log-normal)	min IBS	3
GP_{IBS}	ML (log-normal)	min IBS	3
BP_2	ML (log-normal)	ML with $\alpha = \beta$	2
GP_1	ML (log-normal)	ML with $\mu = 0$ and $\sigma = 1$	1
GP_2	ML (log-normal)	ML with $\mu = 0$	2
GP_3-t	ML (log-normal)	ML	**3**
LP_0	–	–	0
merge	ML (log-normal)	–	0

we also see from the estimated weight ω_{GP}. The realised time-to-hard-freeze in 2018 was in late October.

5. Simulation study

Here, we study the four combination methods in controlled conditions. We strive to mimic our application, so we generate time-to-hard-freeze observations in a single location over a number of years indexed by i. In each year, we have access to two different forecasts from two sources of information: source 1 and source 2. The forecasts are issued as ensembles, with n_1 samples from source 1 and n_2 samples from source 2, from which we estimate single-source survival curves $S_{1,i}(t)$ and $S_{2,i}(t)$ for each year i. Then, we estimate combination parameters across many years.

We compare the performance of a total of 17 different probabilistic forecasts (in the form of survival curves); see Table 1, where all methods are listed. In the first step of the combination procedure, the single-source forecasts may be estimated non-parametrically, using the Kaplan–Meier (KM) estimator, or using maximum likelihood (ML) with the true parametric model (the log-normal distribution; see Section 5.1). In this simulation setup, combining the parametric curves will always be more advantageous (since the true model is known) and we have therefore not pursued the non-parametric option here, except for the HB method, which is always non-parametric.

In the second step of the combination procedure, we investigate two different ways to estimate the combination parameters: maximum likelihood (for LP, GP, and BP), as in Section 3.4.1, or minimising the IBS, as explained in Section 3.4.2.

We include four different versions of the GP method and two versions of BP. The methods denoted by GP_3 and BP_3 in Table 1 are the default versions of the methods described in Section 3.2, each with three combination parameters to estimate. GP_1 and GP_2 are natural simplifications of GP_3: for GP_1, the only parameter left to estimate is ω; the mean is fixed to 0, and the variance is fixed to 1. For GP_2, only the mean parameter is fixed, leaving two combination parameters to estimate (ω and σ). BP_2 is a similar simplification of BP_3, where the beta parameters are constrained to be equal, which fixes the mean of the beta distribution to 1/2. The method denoted by GP_3-t is the Student-t version of the ordinary GP, as presented in Eq. (7).

Finally, we include two natural benchmark forecasts. The first one, LP_0, is simply the linear pool with the combination weight fixed at 0.5. So this method requires no combination parameters to be estimated, and always gives equal weights to the two sources. The second one, 'merge', consists of simply concatenating or merging the two datasets and then fitting a log-normal model to this single, larger dataset.

The complete evaluation results are presented in the appendix, but the results for the methods in bold-face in Table 1 are given particular attention in Section 5.3, below.

5.1. Simulation model

We considered the simulation framework carefully in order to have a suitable benchmark for comparing combination methods in a time-to-event setting. In particular, we took care to construct the framework so that the two sources contain different amounts of information, but where they can both produce calibrated forecasts.

The true realised time-to-hard-freeze in year i is assumed to come from a log-normal distribution, with mean ξ_i and variance τ_0^2 on the log-transformed scale. Here, we let ξ_i be a function of two variance components, $x_{1,i}$ and $x_{2,i}$, which are realisations from normals with mean 0 and variances τ_1^2 and τ_2^2:

$$\text{Truth: } T_i|(x_{1,i}, x_{2,i}) \sim \text{LogNormal}(\xi_i, \tau_0^2)$$
$$\text{with } \xi_i = \xi_0 + x_{1,i} + x_{2,i},$$
$$\text{and } X_{1,i} \sim \text{N}(0, \tau_1^2) \quad X_{2,i} \sim \text{N}(0, \tau_2^2).$$

The random variables $X_{1,i}$ and $X_{2,i}$ reflect some characteristic of each year: for example, if both $x_{1,i}$ and $x_{2,i}$ are

small, there is a high chance of early frost in year i, while if both are large, there is a higher chance of late frost.

For every year i, we generate two sets of forecasts which aim at predicting T_i: n_1 samples from source 1, and n_2 samples from source 2. The data in source 1 are generated with knowledge of $x_{1,i}$, but not of $x_{2,i}$, while source 2 knows $x_{2,i}$, but not $x_{1,i}$. The log-normal has the property that the data from each source are still log-normally distributed:

$$\begin{aligned} &\text{source 1: } Y_{i,j}|x_{1,i} \sim \text{LogNormal}(\xi_0 + x_{1,i}, \tau_0^2 + \tau_2^2), \\ &\text{source 2: } Z_{i,j}|x_{2,i} \sim \text{LogNormal}(\xi_0 + x_{2,i}, \tau_0^2 + \tau_1^2) \end{aligned} \quad (10)$$

for $j = 1, \ldots, n_1$ for source 1, and $j = 1, \ldots, n_2$ for source 2. The relative quality of the two sources depends on the sizes of τ_0, τ_1, and τ_2. Intuitively, if $\tau_2 > \tau_1$ and large compared to τ_0, then source 2 is much more informative than source 1.

Further, we let the two sources have different lead times, which translates into different censoring rates for the samples in the ensembles (see Section 2). In line with our application in Section 6, we let source 1 have a long time, 120 days into the future, leading to a low censoring rate. By contrast, source 2 has a short lead time of 60 days, and therefore a higher censoring rate.

5.2. Scenarios

We consider a total of 16 simulation scenarios; see Table 2. The number of training observations, i.e. the number of past forecast–observation pairs n, can be expected to have a large influence on the estimation of the combination parameters. Scenarios 1 to 8 are idealised scenarios where the combination parameters are estimated based on a large number of training years, $n = 1000$.

The training data consist of realised times to frost, as well as forecasts from the two sources belonging to each realisations. From this training sample, we estimate the combination parameters. Then, we generate a large (10^4) test sample on which the single-source and combined forecasts are evaluated. Scenarios 9 to 16 are more realistic in the sense that we have a small number of training years, $n = 20$, from which the combination parameters are estimated. Each set of combination parameters is then evaluated on a single test dataset, and this whole procedure is repeated a large (10^4) number of times.

The scenarios also differ with respect to the characteristics of the two sources. For some scenarios, we let both sources have small samples, $n_1 = n_2 = 20$, while in the others, we strive to mimic the application in Section 6 and let source 1 have a large ensemble ($n_1 = 100$), while source 2 has a small one ($n_2 = 20$). Note here that source 2 also has a higher censoring rate in all scenarios, so the effective sample size is even smaller. Further, we sometimes let the two sources be balanced in terms of information content, while in other scenarios, we let the sources be very unbalanced. In the first case, we have $\tau_0 = \tau_1 = \tau_2 = 0.4$, and in the second case, we let source 2 have much less information than source 1, with $\tau_0 = 0.53$, $\tau_1 = 0.4$, and $\tau_2 = 0.2$. Finally, we include the possibility of source 2 producing biased forecasts. In these scenarios, we have

$$\text{source 2: } Z_{i,j}|x_{2,i} \sim \text{LogNormal}(\xi_0 + x_{2,i} - 0.5, \tau_0^2 + \tau_1^2),$$

and source 1 as before.

Table 2
Simulation scenarios.

Scenario	No. of training years	Balanced sources	Bias	Ensemble sizes
1	$n = 1000$	yes	no	$n_1 = 100$, $n_2 = 20$
2	$n = 1000$	yes	no	$n_1 = n_2 = 20$
3	$n = 1000$	yes	yes	$n_1 = 100$, $n_2 = 20$
4	$n = 1000$	yes	yes	$n_1 = n_2 = 20$
5	$n = 1000$	no	no	$n_1 = 100$, $n_2 = 20$
6	$n = 1000$	no	no	$n_1 = n_2 = 20$
7	$n = 1000$	no	yes	$n_1 = 100$, $n_2 = 20$
8	$n = 1000$	no	yes	$n_1 = n_2 = 20$
9	$n = 20$	yes	no	$n_1 = 100$, $n_2 = 20$
10	$n = 20$	yes	no	$n_1 = n_2 = 20$
11	$n = 20$	yes	yes	$n_1 = 100$, $n_2 = 20$
12	$n = 20$	yes	yes	$n_1 = n_2 = 20$
13	$n = 20$	no	no	$n_1 = 100$, $n_2 = 20$
14	$n = 20$	no	no	$n_1 = n_2 = 20$
15	$n = 20$	no	yes	$n_1 = 100$, $n_2 = 20$
16	$n = 20$	no	yes	$n_1 = n_2 = 20$

5.3. Results

As stated in Section 3.3, we evaluate forecasts with respect to sharpness under calibration. Calibration is assessed by PIT histograms and by computing the mean and standard deviation of the PIT values. The predictive performance of the forecasts is assessed by the average integrated Brier score (IBS) across all test observations. With the IBS, we use $t_{\max} = 120$ here; see Eq. (8).

We start by presenting the results from the idealised scenarios with a very large training dataset ($n = 1000$), before we present the results for the more realistic scenarios where the training dataset is small ($n = 20$). In the figures below, we present the results for a selection of the methods listed in Table 1. The complete evaluation results are provided in the appendix.

5.3.1. Large training dataset

The left panel of Fig. 5 displays the mean of the PIT values for several methods across all scenarios with the large training dataset. In scenarios where both sources are unbiased, all combination forecasts are unbiased too, i.e. with a mean PIT value close to 0.50. In scenarios where source 2 is biased, we see that the simple combination methods LP and HB tend to produce biased forecasts too, though with less bias than the forecast from source 2.

The right panel of Fig. 4 displays the standard deviation of the PIT values, and demonstrates that the simple combination methods LP and HB fail to produce calibrated forecasts, as expected from the existing literature on forecast combination (see Section 3.2). Even in scenarios without bias, these methods produce overdispersive forecasts, which manifest themselves in PIT standard deviations well below 0.288. The overdispersive forecasts of LP and HB are also clearly visible in the PIT histograms shown in the left panel of Fig. 5

Skill scores (derived from the IBS) with respect to the forecasts from source 1 are displayed in the right

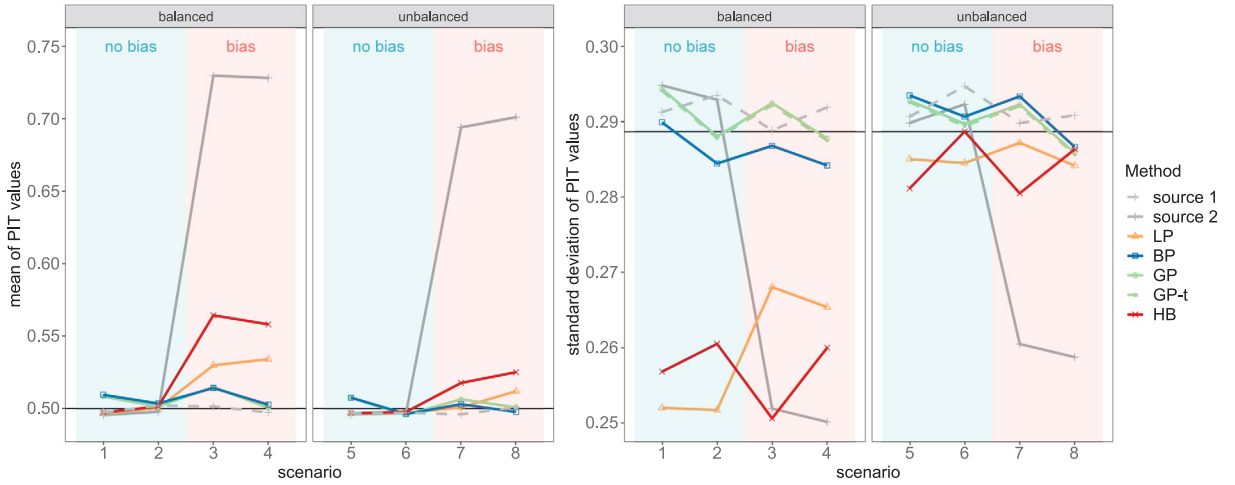


Fig. 4. Mean (left) and standard deviation (right) of PIT values for a selection of methods in the eight scenarios with a large training dataset ($n = 1000$). Perfect calibration is indicated by the straight black lines, at 0.50 for the mean and at $\sqrt{1/12}$ for the standard deviation. Scenarios with bias have a red background, while scenarios without bias have a blue background. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

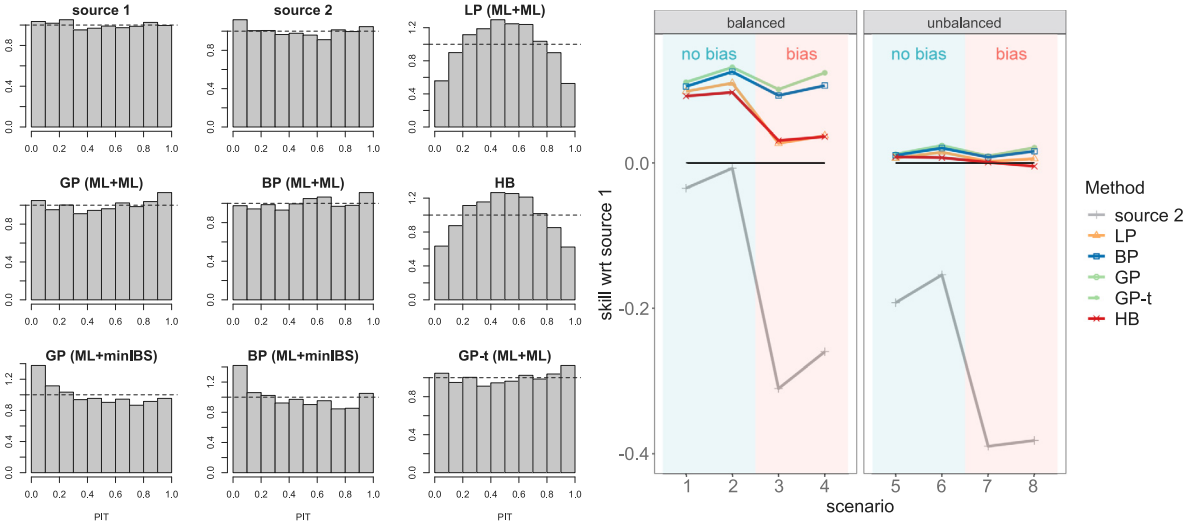


Fig. 5. Left: PIT histograms for scenario 1 (large training dataset, balanced sources, and no bias) for the single-source forecasts and the combined forecasts from methods LP, BP_3, GP_3, HB, BP_{IBS}, GP_{IBS}, and GP_3-t.
Right: IBS skill scores for a selection of methods in the eight scenarios with a large training dataset ($n = 1000$). The straight black line at zero gives the skill of the forecast from source 1: forecasts that lie above the line have better predictive performance than source 1. Scenarios with bias have a red background, while scenarios without bias have a blue background. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

panel of Fig. 4 for a subset of methods. A positive skill score indicates that the forecast has a better predictive performance than source 1. The forecast from source 2 is generally inferior to the forecast from source 1, by design. Source 2 performs on par with source 1 only in scenarios with balanced sources and no bias.

The predictive performance of the combination forecasts is strongly influenced by the degree of imbalance between the sources. When the sources are balanced, all combination methods greatly outperform the single-source forecasts. In these scenarios, the more complex combination methods, BP_3, GP_3, and GP_3-t, also outperform the simpler combination methods (LP and HB), particularly when there is bias in one source. When the sources are unbalanced, the combination forecasts still outperform the single-source forecasts in most scenarios,

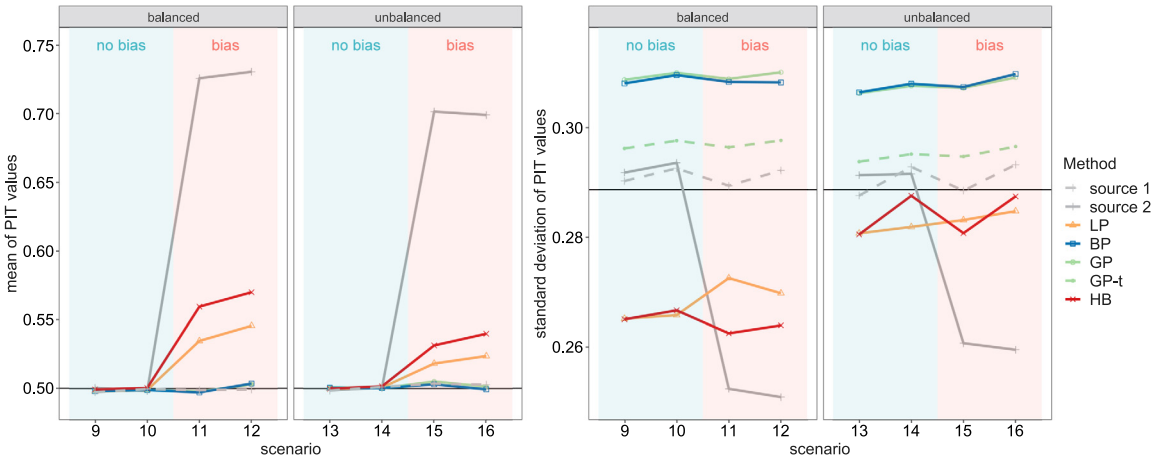


Fig. 6. Mean (left) and standard deviation (right) of PIT values for a selection of methods in the eight scenarios with a small training dataset ($n = 20$). Perfect calibration is indicated by the straight black lines, at 0.50 for the mean and at $\sqrt{1/12}$ for the standard deviation. Scenarios with bias have a red background, while scenarios without bias have a blue background. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

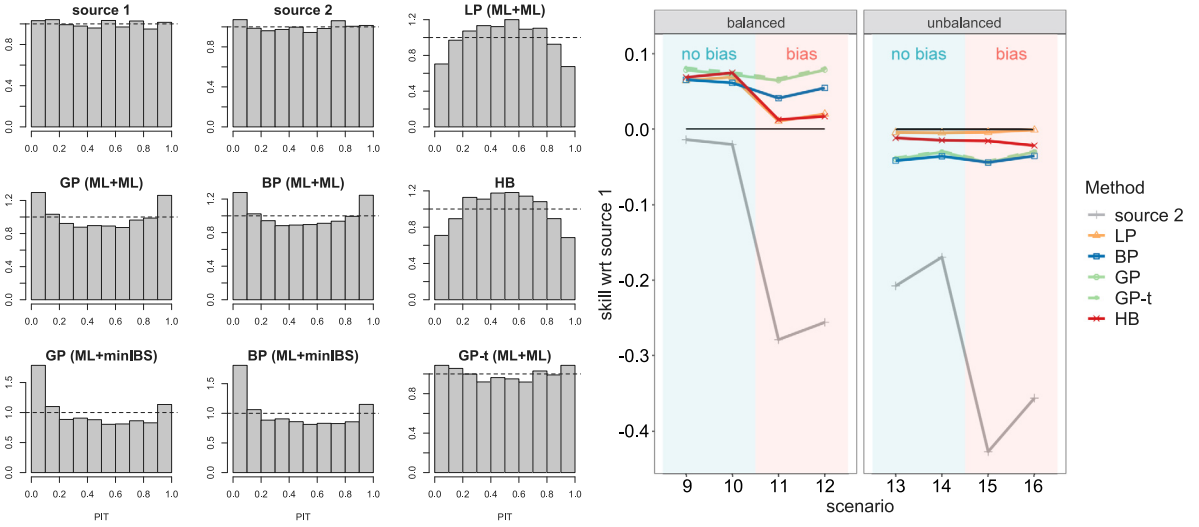


Fig. 7. Left: PIT histograms for scenario 9 (small training dataset, balanced sources, and no bias) for the single-source forecasts and the combined forecasts from methods LP, BP_3, GP_3, HB, BP_{IBS}, GP_{IBS}, and GP_3-t.
Right: IBS skill scores for a selection of methods in the eight scenarios with a small training dataset ($n = 20$). The straight black line at 0 gives the skill of the forecast from source 1: forecasts that lie above the line have better predictive performance than source 1. Scenarios with bias have a red background, while scenarios without bias have a blue background. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

but not considerably. Also, the more complex combination methods only have slightly higher skill scores than the simpler ones.

5.3.2. Small training dataset

Fig. 6 displays the calibration results for several methods in the scenarios with the small training dataset ($n = 20$). The figure with the mean PIT values is roughly similar to the corresponding figure for the large training dataset scenarios. In the figure with the standard deviation of the PIT values (right panel), we see that the forecasts from the simple combination methods LP and HB are still overdispersive, but somewhat less so than in the large n scenarios. Further, the more complex combination methods, BP_3 and GP_3, now produce underdispersive forecasts, with standard deviations clearly above 0.288. In these scenarios, the calibration of the GP forecast is greatly improved by using the Student-t version GP_3-t. This effect is also visible by comparing the PIT histograms for GP_3 and GP_3-t in the left panel of Fig. 7.

The IBS skill scores in the right panel of Fig. 7 demonstrate that when the sources are balanced, all combination forecasts still outperform the single-source forecasts. When the sources are unbalanced, however, all combination methods have worse predictive performance than the forecast from source 1. In those scenarios, the simpler methods are better than the more complex ones, but do not generally beat source 1, nevertheless.

5.3.3. Lessons from simulations

The main finding in these simulations is that combination methods of the type we study here can be beneficial in the sense that they produce forecasts that are calibrated and have higher predictive skill than the single-source forecasts. However, the combination methods require either sufficiently balanced sources or sufficient amounts of training data. In such situations, more complex combination methods will be particularly beneficial, in terms of both skill and calibration. If the sources are unbalanced *and* training data are scarce, all combination methods struggle, particularly the more complex ones.

The simulations also illustrate various other lessons. First, the parametric single-source forecasts outperform the non-parametric ones in terms of predictive performance (see Table A3). This is not surprising, as the parametric single-source forecasts make use of the true model from which the data were generated.

Second, we see that most combination methods generally beat the two benchmark methods in terms of predictive performance and calibration; see the tables in Appendix B. The equal-weight method LP_0 has particularly bad predictive performance in scenarios with bias or unbalanced sources. The naive method, 'merge', fares a bit better, particularly in scenarios with little training data, but is still worse than the proper combination methods in terms of overall predictive performance; see Table A3. Both benchmark methods produce biased forecasts in the scenarios with bias (see Table A1) and overdispersive forecasts in almost all scenarios (see Table A2).

Third, the simulations allow us to compare methods where the combination parameters are estimated by maximum likelihood (ML) and the same methods with combination parameters estimated by minimising the IBS (minIBS). In terms of predictive performance, estimation by minIBS is roughly similar to estimation by ML. For some methods, minIBS is often the best, while for other methods ML is often a bit better; see Table A3. The estimation by minIBS can produce less calibrated forecasts, however; see Figs. 5 and 7, where the PIT histograms for BP_{IBS} and GP_{IBS} reveal some degree of bias. Also, estimation by ML is much faster, particularly for the more complex combination methods.

In addition, we compare different versions of the more complex combination methods, GP and BP; see Table A3. The one-parameter method GP_1 achieves predictive performance on par with the other one-parameter methods LP and HB (even beating both of these in some scenarios). However, it suffers from the same issues as these methods with respect to calibration: overdispersive forecasts in the unbiased scenarios and biased forecasts in the scenarios with bias in source 2. The two-parameter methods GP_2 and BP_2 do not produce overdispersive forecasts (since they can correct the variance), but get biased forecasts in the scenarios with bias. These two methods could then be good choices in applications where the sources are known or assumed to be unbiased. These results illustrate the effect of the extra combination parameter in the more complex combination methods. For GP for example, the mean parameter fixes potential biases and the variance parameter ensures calibration.

Finally, we learn that ensemble size does not tend to have a large effect. At least not for the set of values we studied here.

6. Case study: Time-to-hard-freeze

To test the combination methods in practice, we use a dataset consisting of seasonal and subseasonal temperature forecasts that give information about the first occurrence of the hard freeze after October 1 for a set of locations in Norway and Fennoscandia. Here, a 'hard freeze' refers to a mean daily near-surface air temperature below 0 °C, as described in Section 2. The goal is to test whether a combination of subseasonal and seasonal forecasts gives more accurate time-to-hard-freeze forecasts than any of the two forecasting products alone, and to investigate differences between the combination methods proposed in Section 3.2.

6.1. Data

We use daily temperature data from three different sources: seasonal forecasts initialised on September 1; subseasonal forecasts initialised on September 30; and the actual time-to-hard-freeze observations, which we obtain from the gridded, observation-based data product SeNorge2018 (version 18.12). The forecasts and observations cover the period from 1999 to 2018, which is our study period.

Seasonal forecasts of two-meter temperatures, initialised on September 1, were downloaded from the Copernicus Climate Data Store (CDS)[2] for 1999–2018 and come from the NWP models of five different weather centres (ECMWF, CMCC, DWD, Météo France, and UK Met Office). The number of ensemble members varies between years, but for each year, there are approximately 150 ensemble members in total, with a maximum lead time of five to six months. See Appendix D.1 for more details about the forecasting systems involved.

The subseasonal forecasts are two-meter temperature NWP forecasts delivered by ECMWF, initialised on September 30.[3] We use a forecast produced in 2019, with forecast cycle number CY46R1. Forecasts with issue dates from 1999–2018 were available with 11 ensemble members and a maximum lead time of 46 days (November 15).

The SeNorge product is a dataset developed by the Norwegian Meteorological Institute and contains mean

[2] https://confluence.ecmwf.int/display/CKB/C3S+Seasonal+Forecasts%3A+dataset+documentation

[3] https://confluence.ecmwf.int/display/S2S/ECMWF+model+description

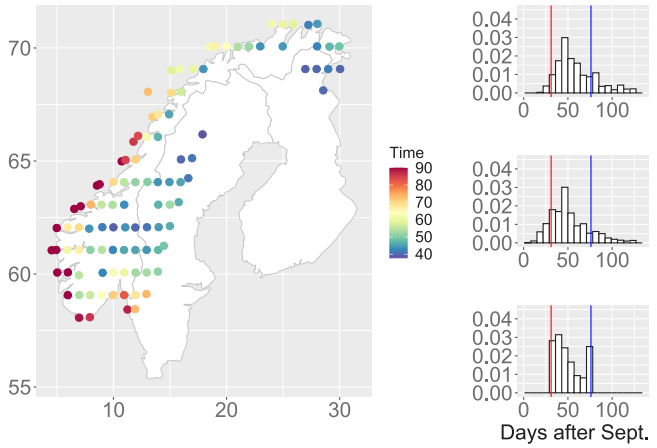


Fig. 8. Left: The 103 study locations in Fennoscandia and their mean observed times to hard freeze from September 1 (1999–2018) according to SeNorge, where the scale is truncated at 122 days (December 31). Right: Distribution of the observed or forecasted time-to-hard-freeze for all locations for 1999–2018 for SeNorge, the post-processed seasonal forecast ensemble, and the post-processed subseasonal forecast ensemble. The sample sizes are 2060, 355,212, and 30,360 for the three data sources. The red line marks October 1 (day 31), while the blue line shows the maximum lead time date of the subseasonal forecast (day 30+46, November 15). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

daily temperature data for Norway and parts of Fennoscandia (Lussana et al., 2018). The product is based on two-meter temperature observations from meteorological stations that are interpolated to a 1 km × 1 km grid by using the optimal interpolation approach (Gandin & Hardin, 1965; Kalnay, 2003). See Lussana et al. (2018) for details. Here, we use the SeNorge data as a substitute for local, observed surface temperatures, as in Roksvåg T. Lenkoski et al. (2023).

Among the large number of grid nodes available in the SeNorge dataset, study locations are selected so that seasonal and subseasonal forecasts are available in reasonable proximity. We only select locations with mean elevation between 0.5 and 800 m a.s.l. The elevation restriction is added because the study is motivated by applications in agriculture and we want to avoid grid nodes that mainly cover the sea, lakes, or mountains. Also, locations that experience the first hard freeze before October 1 for more than five out of our 20 study years are omitted from the analysis. Details of the selection of study locations are given in Appendix D.1. The resulting 103 study locations are shown in Fig. 8, together with histograms of the observed or forecasted times to hard freeze for the three datasets.

Seasonal and subseasonal forecasts are known to be subject to systematic biases (Haiden et al., 2023; Hermanson et al., 2018; Horat & Lerch, 2024; Magnusson et al., 2013), and they come at a considerably coarser resolution compared to SeNorge. To make sure that the forecasted temperatures better match the temperatures observed at the target locations, the raw seasonal and subseasonal forecasts are post-processed in the same manner as in Roksvåg T. Lenkoski et al. (2023), by adjusting the forecast means and standard deviations relative to historical means and standard deviations from SeNorge; see more details in Appendix D.2. Note that there are a number of alternative methods for post-processing that we could have considered, for example, the Bayesian model averaging (BMA) approach (Raftery et al., 2005) or the ensemble model output statistics (EMOS) approach (Gneiting et al., 2005). From the post-processed seasonal and subseasonal temperature forecasts, we extract the day of the first occurrence of the hard freeze after September 1 for each combination of year, location, ensemble member, and forecast system. Likewise, we take out the first day of observed hard freeze for SeNorge.

6.2. Experimental setup

We use a subset of the combination methods from Table 1 to predict the time to the first autumn frost from October 1 by combining forecasts produced from the two data sources:

- Source 1: the seasonal forecast, i.e. an ensemble of around 150 members with an issue date of September 1 and a censoring date of December 31.
- Source 2: the subseasonal forecasts, i.e. an ensemble of 11 members with an issue date of September 30 and a censoring date of November 15.

From the above data sources, we make single-source forecasts by using the non-parametric Kaplan–Meier estimator (KM) or a parametric log-normal distribution for each combination of year and location, independently of other years and locations. We consider October 1 as the issue date of the combined forecast. To account for what happened during the month of September, we remove ensemble members in the seasonal forecast that predict the hard freeze before October 1 if there are any.

After estimating the single-source forecast for a location, we go through each target year (1999–2018) and estimate the parameters of the combination methods based on forecasts from the remaining 19 hindcast years. There are then 19 forecast–observation pairs available for the

Table 3
Mean IBSs for the combination methods (LP, GP, BP, HB, and merge) and the reference single-source forecasts (Clim, Seas9, and Subseas) over all target locations. Models marked 'KM' are based on the non-parametric Kaplan–Meier survival function and estimated by minimizing the IBS, while models with 'ML' are based on the parametric log-normal survival function and fitted using the maximum likelihood estimator. The results for both post-processed and raw forecasts are shown. Combination methods that have significantly lower IBSs than all of the single-source forecasts (Seas9, Subseas, and Clim) according to the Wilcoxon signed-rank test with a significance level 0.05 are marked in bold in each row .

	Clim (KM)	Seas9 (KM)	Subseas (KM)	Seas9 (ML)	Subseas (ML)	LP_0 (ML)	Merge (ML)
Post-pr.	0.0753	0.0758	0.1034	0.0748	0.0854	**0.0747**	**0.0739**
Raw	0.0753	0.0946	0.1396	0.0941	0.1239	0.0918	0.0912

	LP (KM)	LP (ML)	HB (KM)	GP_1 (ML)	BP_2 (ML)	GP_2 (ML)	BP_3 (ML)	GP_3 (ML)
Post-pr.	**0.0733**	**0.0723**	**0.0735**	**0.0726**	**0.0733**	**0.0735**	0.0760	0.0762
Raw	0.0847	0.0837	0.0829	0.0838	0.0808	0.0829	**0.0745**	**0.0772**

estimation of the combination parameters. For this estimation step, we consider both non-parametric and parametric methods; see Section 3.4. In the non-parametric case, with single-source KM forecasts, we test the hazard blending method (HB) and linear pooling (LP). Combination parameters are estimated by minimizing the IBS. In the parametric case, with single-source log-normal forecasts, we fit linear pooling method (LP), Gaussian pooling (GP) with one to three parameters (GP_1, GP_2, and GP_3) and the beta pool (BP) method with two and three parameters (BP_2 and BP_3). Combination parameters are then estimated by maximum likelihood (ML). In addition, we test two benchmark methods: the LP_0 method, i.e. linear pooling with fixed, equal weights on both forecast sources ($\omega = 0.5$); and the 'merge' method where a log-normal model is fitted to a combined ensemble with members from both seasonal and subseasonal forecasts.

The combination forecasts are evaluated by using the IBS from Eq. (8) computed from October 1 (day 31) to December 31 (day 122), with the time-to-hard-freeze from SeNorge as the ground truth. We calculate location-specific IBS', based on the 20 (annual) predictions performed for each location. In the case that the first hard freeze has already occurred at the target location and target year before or on October 1, we set the IBS to 0 for all methods. This happens for 82 of the 2060 location–year combinations (103×20) that we are considering.

The single-source subseasonal and seasonal forecasts are used as reference forecasts, both non-parametric (KM) and parametric (log-normal, ML) versions. In addition, we make a climatology reference forecast by fitting the KM estimator to the historical observed time-to-hard-freeze observations from 1999–2018, with the target year being left out together with years with a hard freeze before October 1 (if any). The results based on both raw and post-processed ensembles are presented.

We use a paired, one-sided Wilcoxon signed-rank test to evaluate whether the IBS from a combination method is significantly lower than the IBS from a reference forecast (single-source forecasts or climatology). The Wilcoxon signed-rank test is a non-parametric test whose null hypothesis states that the distribution of differences between paired data points, in this case, IBSs from a combination method and a reference method, is symmetric around zero (Siegel & Castellan, 1988). Our alternative hypothesis is that the median differences have a mean less than zero. If the null hypothesis is rejected, it suggests that the combination method in question gives significantly lower IBS values than the competing reference method. We perform the test with a significance level of 0.05. Note that significance testing in this setting is an area for future research; one may also consider using permutation tests (Good, 2005) instead of the Wilcoxon signed-rank approach. Permutation tests may be particularly suited to situations with spatial and temporal dependencies.

6.3. Results

The results from the case study are summarized in Table 3 in terms of the location-specific IBSs averaged over all study locations. We see that most combination methods beat the single-source forecasts, and the climatology forecast, when post-processed ensembles are used as input. The simpler combination methods generally perform better than the more complex combination methods. The LP (ML) method has the lowest IBS of all methods, while the methods GP_1, GP_2, and BP_2 have a lower mean IBS' than their more complex counterparts GP_3 and BP_3. This is likely due to the low number of hindcast years (19), which may result in overfitting and unstable parameter estimates, particularly for the methods with several combination parameters.

The best combination methods have somewhat better predictive power than the naive benchmark methods LP_0 and 'merge'. Note also that the log-normal model seems to fit the forecast ensembles well: we see that the predictive performance of the single-source forecasts is actually higher when using the log-normal model (ML) than with the non-parametric estimator (KM). This is the case for both sources, but the effect is particularly large for source 2, which has few ensemble members.

Considering the raw forecasts, the combination methods with three parameters (GP_3 and BP_3) give lower mean IBSs than the simpler methods (GP_1,GP_2, BP_2, HB, and LP). According to Table 3, all the combination methods outperform the single-source forecasts they are based on, but BP_3 and GP_3 also beat the climatology forecast as the only methods, according to the Wilcoxon signed-rank test. The results suggest that the one or two extra parameters in the BP_3 and GP_3 models are more important when we have not already accounted for biases and dispersion through post-processing.

In addition to the combination methods in Table 3, we briefly investigate the spread-adjusted linear pool (SLP)

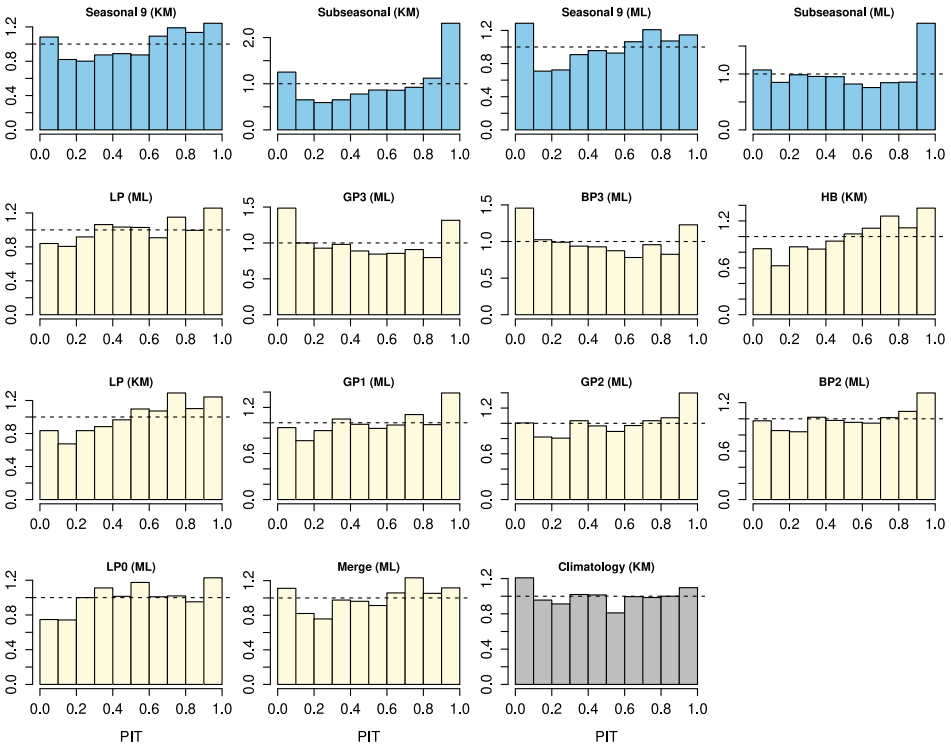


Fig. 9. PIT histograms for the single-source forecasts (blue), combination methods (yellow), and the climatology forecast (gray) for time-to-hard-freeze predictions when post-processed forecasts are used. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

method mentioned in Section 3.2. The SLP method performed better with the post-processed than with the raw forecasts, but did not outperform the best methods in Table 3.

Fig. 9 displays PIT histograms for the time-to-hard-freeze predictions for the combination forecasts and for the single-source forecasts when using post-processed input data. Despite the post-processing, the subseasonal forecasts are clearly biased, with many PIT values around one, suggesting underestimation of the time-to-hard-freeze. The seasonal forecast tends towards overestimation in the ML case. The combination methods based on parametric single-source forecasts (ML) appear to be better calibrated than the methods using non-parametric forecasts (KM). The non-parametric HB (KM) and LP (KM) both show signs of overestimation, while the method that performed best in terms of the IBS, LP (ML), looks well calibrated. See Table A5 for the mean and standard deviation of the PIT values for all methods, and see Figure A1 for PIT histograms for the raw forecasts.

In the remainder of this section, we focus on the best-performing combination method, LP (ML), for the post-processed forecasts. In Fig. 10, the IBSs of LP (ML) are compared to the IBS' of the single-source forecasts. The right plot in the upper panel shows that the IBS of the subseasonal forecast is rather high compared to the climatology forecast for many locations, but there are also locations where the subseasonal forecast gives better predictions than the climatology forecast. Next, considering the right plot in the lower panel, we see how the combination method is able to improve the results at locations where the subseasonal forecast performed poorly, while at the same time keeping the good performance for locations where the subseasonal forecast was performing well.

Fig. 10 further shows that compared to the seasonal forecast, the combination method is better or approximately as good for most locations (lower panel, centre plot). The same applies for the combination method compared to the climatology forecast(lower panel, left plot). We also notice how the climatology forecast and the seasonal forecasts are almost equivalent in performance (upper panel, centre plot), which is in line with the results in Roksvåg T. Lenkoski et al. (2023): the skill of the seasonal time-to-hard-freeze forecast is typically high compared to the climatology forecast for regions where hard freezes occur in the first three weeks after the forecast issue date. In our study, most locations experienced hard freeze on day 31 or later compared to the issue date of the seasonal forecast (see Fig. 8). Currently, the seasonal forecasts we use in this study are only initialised once a month (September 1, October 1, …) and published 12 days after their initialisation date (September 12, October 12, …). Ideally, a seasonal forecast from mid-September, published just before October 1, would be available in real time for combination.

Fig. 11 shows how the skill scores of the LP (ML) method are distributed across our study locations relative

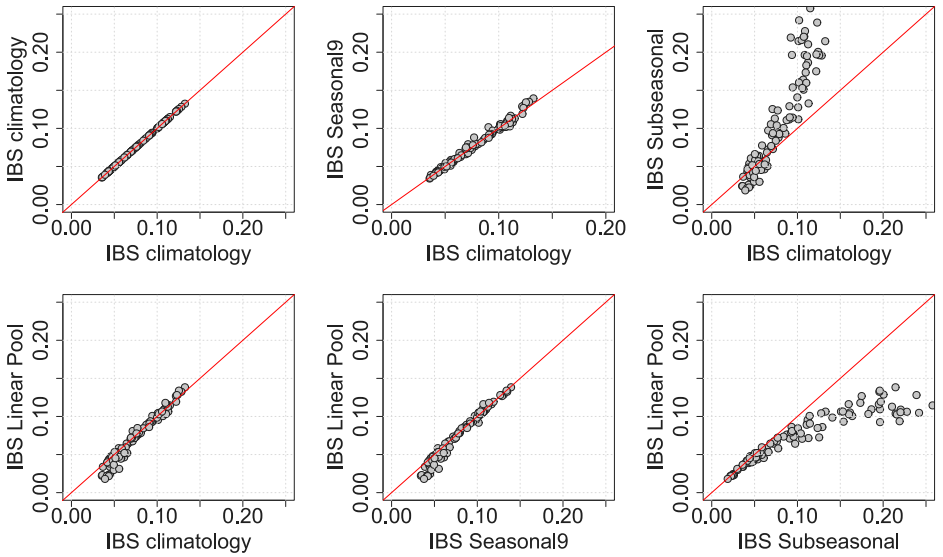


Fig. 10. The top row shows location-specific IBS' for the single-source forecasts, Subseasonal (ML) and Seasonal (ML), compared to the climatology forecast, Clim (KM). The bottom row shows the IBS' for the linear pool combination (ML) method compared to the climatology forecast and the single source forecasts (bottom row). All ensemble forecasts involved are post-processed.

to the subseasonal and seasonal forecasts. The parameter ω from Eq. (4), which determines how much weight is assigned to the seasonal forecast in the LP method, is also displayed. We see that the combination method is considerably better than the subseasonal forecasts in many regions (Fig. 11(b)), but particularly in southern Norway and along the coast. In coastal areas, hard freezes occur late, as shown in Fig. 8, and often after the censoring date of the subseasonal forecast (day 76). It therefore makes sense that the seasonal forecast with a longer lead time is weighted more in these regions, as shown in Fig. 11(c).

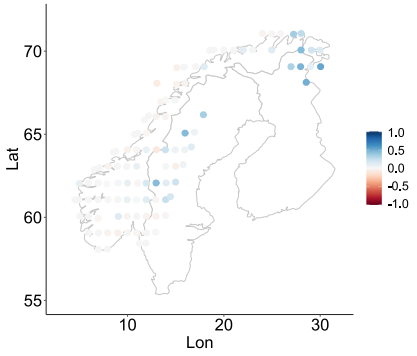
Compared to the seasonal forecast, the LP (ML) method is better in regions where frost comes early, i.e. in the northernmost parts of the study area and in the eastern parts, far away from the coast (Fig. 11(a)). These are regions where the subseasonal forecast performs well according to Fig. 11(b), and the combination method is able to incorporate this by giving the subseasonal forecast a high weight (red areas in Fig. 11(c)). In the remaining areas, the combination method performs similarly to the seasonal forecast (and thus also to the climatology forecast).

7. Discussion and concluding remarks

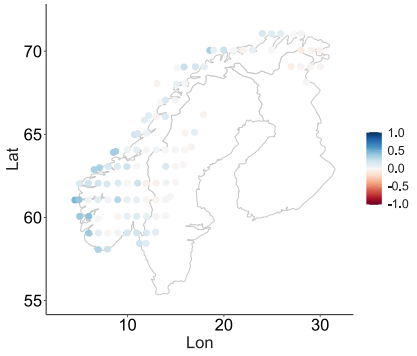
In this paper, we investigated methods for combining forecasts from different sources of information, in particular, the combination of meteorological forecasts for time-to-event outcomes. We introduced two new combination methods, the Gaussian pool and hazard blending, and compared their performance with state-of-the-art methods in an extensive simulation study and in a case study where the goal was to predict the time until the first hard freeze of the season.

In the simulation study, we found that combination forecasts typically outperformed the single-source forecasts they were based on when there were sufficient amounts of training data or when the sources were reasonably balanced. On the other hand, all combination methods struggled when there were little training data and unbalanced sources. In these scenarios, the more complex combination methods were generally worse than the simpler ones. The time-to-hard-freeze case study provided both support and nuance to these findings. In the case study, we had a small training dataset (19 forecast–observation pairs) and we had two sources which were clearly quite unbalanced: the seasonal forecast had a higher predictive performance than the subseasonal forecast, at least when looking at the mean IBS over all locations. From the simulation study, we could therefore have expected that the case study would be a challenging situation for the combination methods. Fortunately, we nonetheless obtained combination forecasts with higher predictive power than the single-source forecasts, and which also outperformed the climatology forecast. Generally, the simpler single-parameter combination methods fared better than the more complex ones, and this was in line with our expectations from the simulation study.

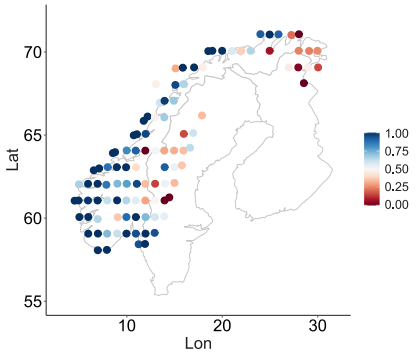
We believe that these findings are likely to hold in the context of combination for other meteorological outcomes, beyond time-to-event setups which we have studied. Similar results have been reported elsewhere, for example by Baran and Lerch (2018), where simpler combination methods generally outperformed the more complex ones in their case studies. In the econometrics literature, however, combination methods generally seem to struggle more than they did in our investigations. Many studies report that estimating combination weights from training data often results in a worse forecast than using fixed, equal weights; see for instance Wang et al. (2022). This phenomenon is referred to as the 'forecast combination puzzle', as discussed by Wang et al. (2022), Zischke et al. (2022), and Frazier et al. (2023). We did not observe this phenomenon: the simple benchmark method



(a) IBS skill score relative to the seasonal forecast (and climatology).



(b) IBS skill score relative to the subseasonal forecast.



(c) Weight ω on the seasonal forecast in the LP (ML) method.

Fig. 11. Location-specific skill scores of the LP (ML) combination forecast relative to the seasonal (Fig. 11(a)) and subseasonal forecasts (Fig. 11(b)) when post-processed forecasts are considered. As the seasonal forecast and the climatology forecast are similar in terms of the IBS (see Fig. 10), the skill scores of LP compared to the climatology forecast (Clim, KM) are almost identical to what is shown in Fig. 11(a). Fig. 11(c) presents the median weight parameter ω for each location, and shows how much weight is put on the seasonal forecast in the LP (ML) method.

LP_0 with fixed, equal weights resulted in lower predictive performance than most other combination methods in both the simulation study and the case study. As mentioned in the introduction, the reason might be that forecasts in meteorology are fundamentally different from forecasts in economics, for instance. In particular, the forecasts in meteorology originate from physical models rather than statistical models on historical data, and this likely makes meteorology a field where combination methods should be particularly fruitful.

Gneiting and Ranjan (2013) argue that combination methods ought to be flexibly dispersive, so that they produce calibrated forecasts. To achieve this property, more combination parameters are required, and we see that in practice, the fruitfulness of such approaches depends on the size of the available training dataset and on aspects of the sources that we are combining. Still, we certainly think that the more complex, flexibly dispersive methods have a role to play, particularly in situations where the data sources are reasonably balanced or when there are sufficient training data. Also, we can consider more complex estimation schemes for the combination parameters, which could reduce the risk of overfitting to the limited training dataset. One may for example use regularisation methods in the estimation, as considered by Diebold et al. (2022). Also, more complex modelling of the combination parameters across time and space, for example, as is often done in econometrics; see for instance Aastveit et al. (2022) and Wang et al. (2022). Specifically, for our application, one could have considered letting some or all combination parameters depend on spatial covariates like elevation or spatial random effects (i.e. assuming some degree of similarity in the parameters that belong to locations that are close). This would have allowed us to borrow strength across locations and would likely have resulted in more stable estimations of the combination parameters. We plan to pursue ideas like these in future work.

CRediT authorship contribution statement

Céline Cunen: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Thea Roksvåg:** Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation. **Claudio Heinrich-Mertsching:** Writing – original draft, Methodology, Formal analysis. **Alex Lenkoski:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Thea Roksvaag, Alex Lenkoski, Claudio Heinrich Mertsching reports financial support was provided by Research Council of Norway. Thea Roksvaag, Alex Lenkoski, Claudio Heinrich Mertsching reports financial support was provided by European Commission. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The data and code used in this manuscript are available on GitHub at https://github.com/NorskRegnesentral/CombiningTimeToEvent. Instructions for reproducing the results are included in the repository's README file.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.10.004.

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